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Course: Data handling and
Visualization

Code: DSA0607

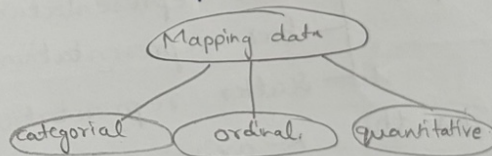
① Concept of Mapping data and types of data

Mapping data:-

Mapping data is used to convert the elements to the visual representation by using position, size, colour and shape.

Types of data:-

- ① Categorical data
- ② ordinal data
- ③ quantitative data



Categorical data - This data represents groups or categories rather than numerical values. It is used to clarify or label items.

Ex:- Gender, Blood group etc..

ordinal data - This is consisting of numbers that represents measurable quantities.

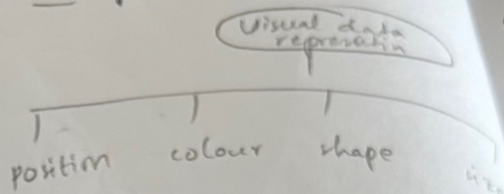
Ex:-

quantitative data - The data which contain the ordinal information

- a) Discrete data
- b) Continuous data

Visual Representation aspects of data

- ① Position
- ② colour
- ③ shape
- ④ size



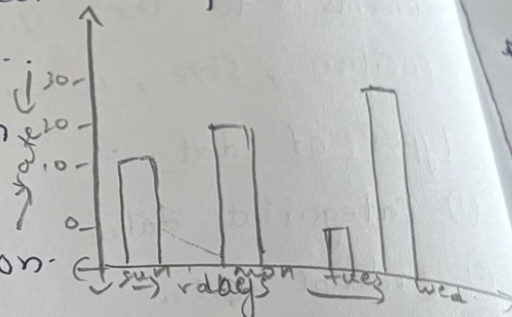
⇒ position:-

This is the visual representation by the graphs where number is not needed by that height we can check the position.

Ex Ranking, graphs ect.

Graphs

- line representation
- bar representation
- scatter representation



* In this x-axis, y-axis or horizontal, vertical axes are needed.

⇒ colour:-

By the colour we can predict faster whether it's an emergency or not, like the colour is an alert for representation.

Ex used in pie charts, Graphs.

① shapes:-

By the shapes it is used to predict which is higher in place or population.

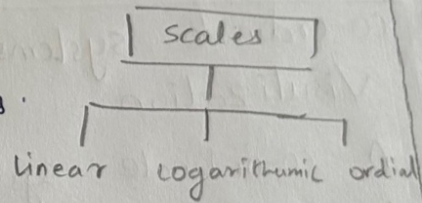
Representation 2 In pie charts, Bubble representation

Size This is the representation of countries, in case of growth, population ect..
Color Bubble representation.

Role of scales in data visualization

A scales in data visualization is a rule function that maps data value to visual properties such as position, length, size, colour or shapes.

- * Convert data into visual form
- * Ensures data accuracy representation
- * Improve Reliability
- * Enable comparison
- * Highlight trends and patterns.



Types of scales:

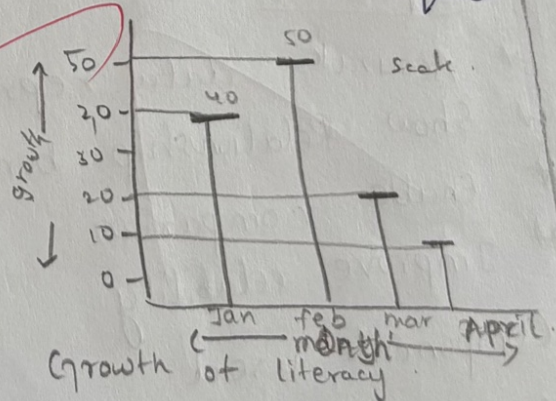
① Linear scale:

It maps equal difference, in data to equal distance on the graph.

eg - 0, 20, 40, 60, 80

Applications

Bar graph, line graph



② Logarithmic scale

It represents values based on their ratios than their absolute difference

eg population growth, earthquake.

Applications:-
Financial growth.

② Ordinal scale:-

The scale that represents categories that have a magnifying order but not equal spacing have

Ex poor $\rightarrow$ fair $\rightarrow$ good $\rightarrow$ excellent.

Application:-

- * Survey result
- * Educational levels.

Impact

- * Hides importances
- * creates False impression
- * Reduces accuracy.

③ Co-ordinate systems and Axes in Data Visualization.

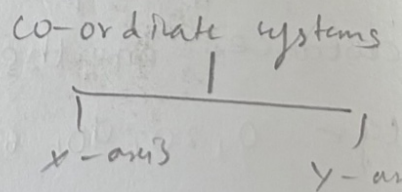
A co-ordinate system is a framework used to locate and represent data points in a group or visualization.

Importance

- * Accurate data representation.
- * Show Relationship
- * Enable comparison
- * Improve reliability

Components used

X-axis $\rightarrow$ used to place horizontal things
Y-axis $\rightarrow$ used to place vertical things.



origin - The point which divides in to neutral.
Cartesian co-ordinate system

Every data points represented by an
ordered pair (x, y)

- * Two perpendicular axes
- * enable spacing
- * suitable for numerical and continuous data
- * easy to compute exact values.

Eg: Bar graphs, Area charts.

② polar co-ordinates

Instead of origin x and y each
point is written as r, θ

- * circular layout
- * Measures distance and angle
- * Best for directional, cycle.
- * suitable for displacing.

Eg: Radar charts
polar charts
circular plots

- * A weather station displays wind direction and speed using polar chart.